

Literature Research on Communication Methods of Multi-Agent Reinforcement Learning in Traffic Signal Control

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Abstract—Efficient traffic signal control (TSC) is essential for mitigating urban congestion and enhancing environmental sustainability. While Deep Reinforcement Learning (DRL) has emerged as a powerful paradigm for adaptive signal timing, single-agent approaches often fail to address the complex spatio-temporal dependencies of large-scale urban grids. Multi-Agent Reinforcement Learning (MARL) provides a collaborative framework, yet achieving network-wide optimization relies heavily on effective communication methods between agents. This paper provides a comparative literature survey of communication-based MARL in the TSC domain, analyzing one foundational review and four representative research articles. We first establish a unified Markov Decision Process (MDP) formulation for TSC and categorize contemporary communication methods into three distinct paradigms, analyzing their main contributions. Furthermore, we conduct a cross-platform evaluation focusing on control performance, scalability and efficiency, and policy transferability to synthesize experimental outcomes across varying simulation environments. Finally, we identify critical open challenges and propose future research directions for developing more robust and transferable multi-agent coordination strategies in intelligent transportation systems.

Index Terms—Intelligent transportation systems (ITS), Traffic signal control (TSC), Multi-agent reinforcement learning (MARL), Agent communication, Graph neural network

I. INTRODUCTION

Traffic congestion is a serious problem to solve as it affects not only travel efficiency but also the environment. Traffic signal control (TSC) is one of the aspects that could reduce congestion. It aims to control signal timing to improve the vehicles' passing efficiency [1], thereby providing safe, effective and reliable transportation systems to participants [2].

Historically, TSC research transitioned from pre-timed control to actuated and adaptive methods like SCOOT or SCATS. However, these traditional approaches often rely on simplified traffic flow models that struggle with the stochastic and non-linear nature of modern urban traffic [3]. Unlike supervised learning or unsupervised learning, Reinforcement Learning (RL) operates as a distinct machine learning paradigm designed for sequential decision-making. By interacting with the environment through a trial-and-error process rooted in the Markov Decision Process (MDP), RL can derive optimal control policies based on rewarding or penalizing criteria. When combined with the function approximation power of deep learning, the resulting Deep RL (DRL) framework is currently considered as the state-of-the-art learning framework in control systems [2].

The TSC problem is naturally modeled as a Markov Decision Process (MDP), where an agent learns to map traffic states to signal actions to maximize cumulative rewards (e.g., minimized delay). While single-agent RL can optimize isolated intersections, it suffers from a “narrow vision” in urban grids where the arrival of vehicles at one junction is directly dictated by the signal timing of its neighbors. Multi-Agent Reinforcement Learning (MARL) addresses this by treating each intersection as an individual agent. Through coordination and communication, these agents can move beyond local optima to reach an equilibrium or optimum policy across the entire network, effectively managing the spatio-temporal dependencies of traffic flow.

Notably, because real-world experimentation with untrained RL policies poses safety risks, the field relies almost exclusively on high-fidelity simulation environments [2]. Platforms such as Simulation of Urban MObility (SUMO) and CityFlow have become the industry standard, allowing researchers to create digital twins of real-world maps and traffic datasets. This simulation-first approach provides the necessary “playground” for the complex multi-agent architectures discussed in this study.

The goal of this article is to analyze and compare the MARL communication methods in the TSC area. The five selected papers include a comprehensive survey on intelligent transportation systems [2] and four research papers representing distinct innovations in multi-agent communication [1], [4]–[6]. Our analysis follows a structured progression: first, we formulate the TSC problem as an MDP and summarize the states, actions, and rewards defined across the five papers. Next, we categorize the communication methods into three types and detail their respective model architectures and RL methods, analyzing how they address specific traffic challenges. In the experimental results and comparisons section, we evaluate the models across four papers based on control performance, scalability and efficiency, and policy transferability, enabling a cross-platform comparison. Finally, we synthesize the findings to identify open research challenges and propose future directions.

II. PROBLEM FORMULATION

This section formulates the multi-agent traffic signal control problem within the Markov Decision Process (MDP) framework. Unlike single-agent scenarios, MATSC requires coordi-

nating multiple autonomous controllers to optimize network-wide traffic efficiency. We first introduce the general MDP formulation and its key components — state, action, and reward — based on the foundational taxonomies provided in the literature survey [2]. Following this, we analyze how the four selected studies instantiate these components, focusing on their specific motivations such as spatial-temporal correlation and communication efficiency.

A. Markov Decision Process Formulation

Following the standard formulation in Deep Reinforcement Learning for Intelligent Transportation Systems: A Survey [2], we model the control problem as a Markov Decision Process (MDP) defined by the tuple $\langle S, A, T, R, \gamma \rangle$, where:

- S : state space (observable traffic conditions at intersections),
- A : action space (signal phase selection or duration control),
- $T(s_{t+1}|s_t, a_t)$: transition probability to next state,
- $R(s_t, a_t, s_{t+1})$: immediate reward (e.g., waiting time reduction),
- $\gamma \in [0, 1]$: discount factor controlling the weight of future rewards

The Markov property holds: s_{t+1} depends only on (s_t, a_t) , not on the previous state $(s_0, s_1, \dots, s_{t-1})$. The objective is to find an optimal policy π^* maximizing the expected discounted cumulative reward:

$$\mathbb{E} \left[\sum_{i=0}^{T-1} \gamma^i r_{t+i} \right]. \quad (1)$$

Training episodes terminate at a fixed time horizon T or upon reaching a terminal state s_T .

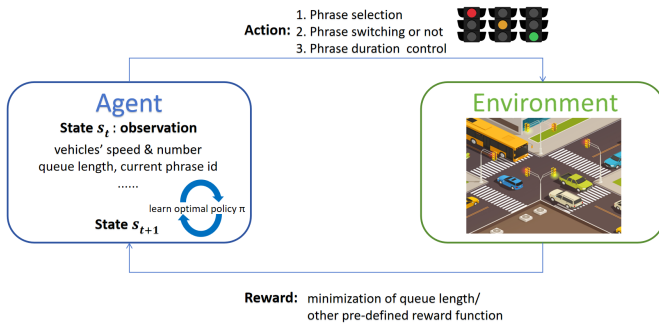


Fig. 1. Markov Decision Process in Reinforcement Learning

B. State

The definition of state (or observation) has a great impact on the effect of reinforcement learning. The review points out that there are two popular types of state representation: vehicle-based discrete traffic state encoding (DTSE) matrix and feature-based vector. While DTSE provides high-resolution, image-like grids capturing precise vehicle positions and speeds, feature-based vectors—which aggregate lane-level metrics like average speed or vehicle counts—are more prevalent in practical applications due to their lower computational

overhead. In addition, neighboring traffic light information might also be included in state definitions.

Given that a single agent typically lacks a global view of the entire network, TSC is often framed as a Partially Observable Markov Decision Process (POMDP). Consequently, the input to the agent is described more accurately as an observation rather than a full state. The four selected studies exhibit a clear progression in how they construct observations to mitigate the limitations of partial observability:

- **Local-centric Representation (QRC-TSC):** The observation of QRC-TSC comprises data from sensors with a 50-meter detection range, including local vehicle counts n , normalized speeds s , queue lengths q and current phase IDs. This design is intended to leverage standard practical sensors to collect essential traffic information.
- **Graph-based Neighborhood Integration (CoLight & Decentralized GNN):** These two studies share a similar logic by moving beyond purely local observations.
- **CoLight:** The observation of each agent consists of the current phase and the number of vehicles on each lane. Each agent can observe part of the system state since it employs a Graph Attentional Network (GAT) to dynamically weight features from neighboring intersections, effectively building an “attentional neighborhood” state. The neighborhood scope could be defined through geographic road distance or node distance (smallest hop count).
- **Decentralized GNN:** Nodes’ state set contains each lane’s normalized queue length, inflow, and mean speed. This research further advances the state representation using a bi-directional graph representation to estimate the state transition.

Both CoLight and the decentralized GNN model adopt the *feature-based vector* representation. CoLight’s state consists of a one-hot encoded phase and the number of vehicles on each lane (aggregated lane-level values), while the decentralized GNN model uses normalized queue length, inflow, and average speed per link. Both approaches aggregate traffic information at the lane or link level, making them feature-based vector methods. Additionally, both models incorporate neighboring traffic information through graph structures or attention mechanisms. A comparison between their graph construction is shown in Table I.

- **Hierarchical Group-based State (CGB-MATSC):** Distinct from the others, this study leverages a k -nearest-neighbor joint state. Each agent has a local state, and each local state abstracts discrete congestion levels $\{0, 1, 2\}$ from the sum of vehicle counts in this intersection, indices of the two most congested traffic light intersections, their respective congestion levels and previous action. Then they integrate k local states as a joint state for one agent.

C. Action

After receiving the state, the agent takes actions from a set of possible actions. The review identifies three primary

TABLE I
COMPARISON OF GRAPH CONSTRUCTION BETWEEN COLIGHT AND
DECENTRALIZED GNN.

Aspect	CoLight	Decentralized GNN
Nodes	Intersections	Traffic links (cells)
Edges	No explicit edge features; neighborhood defined by distance/hops	Bidirectional edges representing turning movements
Node features	Current phase (one-hot), vehicle counts per lane	Queue length, inflow, speed (normalized)
Edge features	None (attention weights are computed dynamically)	Saturation flow, turning ratio
Neighborhood definition	Top- K closest intersections (geo-distance or hop count)	Naturally defined by link connectivity

types: multiple choices (choosing one of the possible green phases), binary choices (switching or staying the phase), and continuous output (adjusting the predefined phase duration).

- **Discrete Phase Selection (QRC-TSC & CoLight):** Both models adopt the “multiple choices” paradigm. Agents choose the next target green phase from a pre-defined set at each interval. This design logic simplifies the decision process into a classification task, which is robust for complex intersections with numerous phase combinations.
- **Hybrid Strategy Selection (CGB-MATSC):** This study uses a unique binary choice logic different from simply choosing switch the phase or not. The agent chooses between two high-level control strategies: a pheromone-based green-wave control or a fixed periodic control, aiming to balance regional stability with local flexibility.

$$\text{Action: } \begin{cases} a^i = 0 : \text{green-wave generation} \\ a^i = 1 : \text{periodic control} \end{cases}$$

- **Continuous Phase Splitting (Decentralized GNN):** In contrast, this study utilizes a continuous action space to output “green time ratios”. The motivation is to overcome the exponential growth of action spaces in multi-phase settings, allowing for finer-grained control over cycle splits compared to discrete switching.

D. Reward

The reward function guides the agent’s policy toward the optimization goal, respond to the agent’s action with reward or penalization. It typically involves the minimization of travel delay or queue length.

- **Queue-based Penalization (CoLight & QRC-TSC):** Both models utilize the negative sum of queue lengths (or halted vehicles) as the primary reward. The design logic is that queue length is a simply, direct, easy-to-measure proxy for intersection delay. For **CoLight**, the reward is the negative sum of queue lengths:

$$r_i^t = - \sum_l u_{i,l}^t, \quad (2)$$

where $u_{i,l}^t$ is the queue length (number of vehicles) on lane l controlled by agent i at time t . Notably, **QRC-**

TSC incorporates a communication-aware regularization term to penalize excessive messaging, balancing traffic performance with communication overhead.

- **Throughput and Efficiency Maximization (Decentralized GNN):** This study defines reward as the “service rate” (ratio of passing vehicles to total demand):

$$r_i(t) = \sum_{s=1}^{|P|} \sum_{(j,k) \in P_s^i} \frac{T_{j,k}(t)}{(w_j(t) \cdot l_j(k) + q_j^{\text{in}}(t)) \cdot r_{j,k}}, \quad (3)$$

where $T_{j,k}(t)$ is the number of turning vehicles from link j to link k during interval $\Delta t = c_i$, $w_j(t)$ is the maximum queue length per lane of link j , $l_j(k)$ is the number of lanes dedicated to the turning movement (j, k) , $q_j^{\text{in}}(t)$ is the inflow of link j , and $r_{j,k}$ is the fixed turning ratio. Unlike simple queue minimization, this design aims to prevent “gridlock” by ensuring that the intersection maintains a high throughput relative to the incoming traffic pressure.

- **Spatially Discounted Joint Reward (CGB-MATSC):** Similar to the design logic in CoLight’s state representation, this model incorporates a spatial discount factor $(1/d_{ij})$ into its reward. An agent’s reward is a sum of its local performance and its weighted neighbors’ feedback, explicitly encouraging cooperative behavior across the group.

III. KEY RL METHODS

This section reviews the core reinforcement learning techniques used in the four selected studies and examines how communication is incorporated into policy learning. In multi-agent traffic signal control, decisions at one intersection can influence queues and arrivals at nearby intersections. Thus, communication can provide additional traffic information for value estimation or policy optimization, helping agents learn more coordinated control policies.

Based on how communication supports coordination, these models can be organized from local information selection, to structural message propagation, and then to regional control. At the local level, attention-based or selective communication focuses on how an agent uses information from nearby agents, either by weighting relevant neighbors or by learning whether specific messages should be transmitted. At the structural level, GNN-based communication embeds information exchange into the road-network graph and updates graph elements through message passing. At the regional level, group-based communication moves coordination from individual intersections to larger control units managed by higher-level regional agents. Table II summarizes the four selected models under this classification.

A. Integrated Architecture of Selected Models

1) **CoLight: Attention-Based Communication with Graph Attention Network (GAT):** CoLight addresses the problem that existing MARL methods either neglect dynamic neighbor influence or rely on fixed neighbor indexing, causing conflicting

TABLE II
OVERVIEW OF SELECTED MARL COMMUNICATION METHODS FOR TRAFFIC SIGNAL CONTROL

Method	Communication Paradigm	Coordination Level	Core Communication Design	RL / Policy Learning Design
CoLight	Attention-based	Local intersection-neighbor level	GAT-based neighbor weighting; index-free aggregation	DQN; parameter sharing; discrete phase selection
QRC-TSC	Selective	Local intersection-intersection level	Learned message mask; bi-level selective communication	Q-MIX; CTDE; joint traffic-control and communication learning
Decentralized GNN	GNN-based	Structural graph-representation level	Bidirectional graph; link-level message passing	DDPG; CTDE; continuous phase-split control
CGB-MATSC	Group-based	Regional group-agent level	CVIS hierarchy; MEC regional aggregation; green-wave information	CGB-MAQL; Q-Learning-Style Regional Value Learning; spatial reward

learning when agents share parameters. In this framework, each intersection is an independent agent, and all agents share the same network parameters to enable scalability to hundreds of intersections.

The observation for each agent consists of its current phase and per-lane vehicle counts; the reward is the negative sum of queue lengths. CoLight adopts Deep Q-Network (DQN) as the core RL algorithm, with all agents sharing the same Q-network parameters. To incorporate spatial cooperation, CoLight augments DQN with Graph Attention Network (GAT) layers: the GAT dynamically computes attention weights between neighboring intersections based on their hidden states, then aggregates neighbor embeddings in an index-free weighted sum. This allows an agent to selectively focus on the most influential neighbors (e.g., upstream intersections during rush hours) without conflicts from fixed ordering.

The resulting cooperatively enhanced representation is used for Q-value prediction, and each agent outputs a discrete action selecting a traffic phase from a predefined set for the next control interval.

2) *QRC-TSC: Selective Communication with Value Decomposition*: QRC-TSC focuses on the communication overhead caused by continuous message exchange among neighboring agents, where redundant information may increase bandwidth usage and add noise to policy learning. To solve this problem, it treats communication as a learnable decision rather than a fixed full-communication assumption.

The model follows a value-based MARL framework under Centralized Training and Decentralized Execution (CTDE). Q-MIX decomposes the global action-value into individual agent utilities, enabling decentralized traffic-control decisions after training. At the same time, each agent learns a communication policy that applies a message mask for each receiver. The mask selects useful message bits, while the receiver-specific design decides which neighboring agent should obtain them.

Thus, communication is integrated into policy learning together with traffic-control decisions. A mutual-information-based objective encourages transmitted messages to be useful for the receiver's action selection, allowing QRC-TSC to reduce redundant communication while maintaining coordi-

nation.

3) *Decentralized GNN: Graph Neural Network-based Communication Using Bidirectional Graph*: Decentralized GNN addresses the problem that existing MARL methods ignore link-level spatial correlations and fail to produce policies transferable to unseen demand patterns. In this framework, each traffic signal is an agent, and agents are decentralized, sharing no parameters. The model is trained under the CTDE paradigm.

The state is represented as a bidirectional graph where nodes are traffic links and edges encode turning movements and saturation flows. A message-passing GNN iteratively updates node and edge features over multiple hops, producing a contextualized embedding for each agent. The model adopts Deep Deterministic Policy Gradient (DDPG) to handle the continuous action space (green time ratios for a fixed-cycle, four-phase system). The critic network is trained centrally using global information, while the actor network conditions only on local graph embeddings. Coordination is achieved through multi-hop message propagation across the graph, enabling agents to sense traffic conditions several intersections away without explicit attention weights.

Each agent outputs a continuous action vector specifying the green time ratio for each phase, which is then converted to actual phase durations. This output directly determines the phase split for the next cycle.

4) *CGB-MATSC: Regional Agent Abstraction and Value-based Coordination*: CGB-MATSC is designed for the scalability challenge in large-scale traffic signal control by changing the basic unit of decision-making. Instead of treating each intersection as an independent RL agent, the model groups multiple traffic lights into one regional unit controlled by a Mobile Edge Computing (MEC) server. In this way, one agent manages a group of intersections, which reduces the number of learning agents and limits the growth of the joint state-action space in large road networks.

The model is built on a hierarchical Cooperative Vehicle-Infrastructure System (CVIS) architecture, where vehicles, Road Side Units (RSUs), and Multi-access Edge Computing (MEC) servers work together to collect and transmit regional

traffic information. Vehicles and RSUs provide real-time traffic conditions, while the MEC server aggregates this information and decides how the traffic lights in its region should be controlled. Therefore, communication in CGB-MATSC is mainly used to support regional-level information collection and coordination, rather than direct message selection between individual intersection agents.

A key part of this architecture is pheromone-based regional green-wave control. Green-wave control means coordinating a sequence of traffic lights so that vehicles can pass through several intersections with fewer stops. In CGB-MATSC, pheromone information is used to identify heavily used or congested traffic directions, and the regional agent can choose a green-wave control mode to give smoother passage to important traffic flows. The agent does not simply choose one signal phase for one intersection; instead, it chooses between high-level control strategies, such as green-wave generation and periodic control.

The RL component is implemented through CGB-MAQL, a value-based MARL method. Each regional agent uses a k-nearest-neighbor-based state to include nearby regional traffic conditions, and its reward combines local performance with spatially discounted feedback from neighboring regions. This reward design encourages the agent to consider not only its own region but also the traffic influence on surrounding areas. As a result, CGB-MATSC connects RL and communication through a regional control pipeline: traffic information is collected through the CVIS structure, the MEC agent selects a regional control strategy, and the spatial reward guides neighboring regions toward cooperative traffic optimization.

B. Cross-model Comparison of Policy Learning

Policy learning explains how each model updates its control strategy from interactions with the traffic environment. In traffic signal control, the choice of policy learning method is closely related to the action space. For discrete signal-phase or strategy-selection problems, value-based methods are commonly used because they can compare candidate actions through estimated Q-values. For continuous control problems, such as optimizing green time ratios, standard DQN-style methods are less suitable because they rely on selecting from a finite action set. Therefore, continuous-action models usually require policy-gradient or actor-critic methods that can directly output continuous control values.

Because the selected models define their actions and coordination structures differently, their policy learning methods also differ.

1) *CoLight: DQN-based Value Learning with Graph Attention*: CoLight formulates traffic signal control as a discrete phase-selection problem. Each agent selects one phase from a predefined action set, so the policy can be learned by estimating the Q-value of each candidate phase. The basic DQN update is based on the temporal-difference target:

$$y_i^t = r_i^t + \gamma \max_{a'} Q(o_i^{t+1}, a'; \theta^-), \quad (4)$$

where o_i^t is the observation of intersection i , r_i^t is the immediate reward (2), and θ^- denotes the target network parameters.

The Q-network is trained by minimizing the difference between the predicted Q-value and the TD target:

$$L(\theta) = \mathbb{E} \left[(y_i^t - Q(o_i^t, a_i^t; \theta))^2 \right]. \quad (5)$$

In CoLight, the Q-value is not predicted from the local observation alone. The observation is first embedded and then passed through graph attention layers, so the final Q-value prediction uses dynamically weighted neighboring information:

$$\hat{q}(o_i^t) = GAT(o_i^t, \mathcal{N}_i) W_p + b_p. \quad (6)$$

Here, $\mathcal{N}_i$ denotes the neighborhood of agent i . This formulation avoids listing all intermediate embedding equations while still showing the key idea: graph attention modifies the state representation before Q-value prediction.

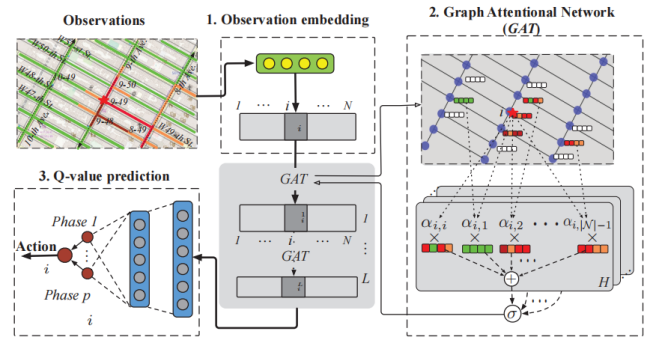


Fig. 2. CoLight Mechanism

This method is suitable for CoLight because the action space is discrete and the model needs to select the best phase at each intersection. DQN directly supports discrete action selection, while graph attention allows the Q-value estimation to reflect the changing influence of neighboring intersections. Parameter sharing further improves scalability because all intersections can use the same policy network instead of training one separate model for each signal.

2) *QRC-TSC: Q-MIX Value Decomposition with Learnable Communication*: QRC-TSC adopts discrete signal phase selection and uses Q-MIX to connect individual action selection with cooperative value learning. In a multi-intersection setting, directly optimizing the joint action of all agents becomes difficult because the joint action space grows rapidly with the number of intersections. As a result, centralized joint-action learning is computationally expensive and difficult to scale.

To address this problem, QRC-TSC uses the Q-MIX-based structure shown in Figure 3, which learns a centralized global action-value function while decomposing it into local utility functions:

$$Q_{tot} = f(Q_1, \dots, Q_n), \quad \frac{\partial Q_{tot}}{\partial Q_i} \geq 0. \quad (7)$$

Here, Q_i is the local utility of agent i , and Q_{tot} is the global action-value. The monotonicity constraint means that

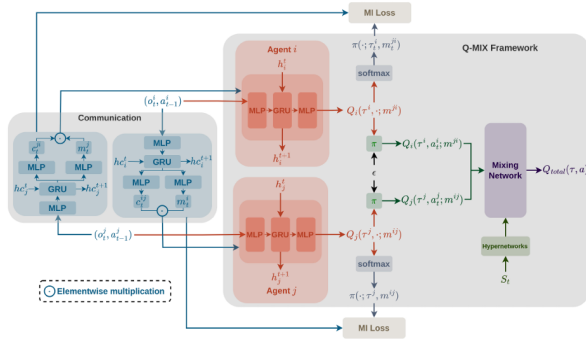


Fig. 3. QRC-TSC Mechanism

increasing a local utility will not decrease the global value. Therefore, the greedy action selected from each local utility is consistent with the greedy joint action under the global value function. This property allows QRC-TSC to use centralized value learning during training while keeping decentralized action selection during execution.

Communication enters this value-learning process through receiver-specific message selection. For each potential receiver j , sender i generates a message m^{ij} and a communication action c^{ij} , where c^{ij} serves as a bit-level mask:

$$\hat{m}^{ij} = m^{ij} \odot c^{ij}. \quad (8)$$

The mask selects which message bits are transmitted to receiver j , so the model learns both message content selection and recipient-specific transmission. These selected messages are incorporated into the receiver's utility network and directly affect the local utility Q_j mixed into Q_{tot} during Q-MIX training. To train the mask, QRC-TSC uses a mutual-information-based objective, encouraging the transmitted message to reduce uncertainty about the receiver's policy or action choice. A compression term further discourages unnecessary dependence on the sender's full action-observation history, so the learned messages remain compact and useful.

The overall objective combines the Q-MIX temporal-difference loss and the mutual-information-based communication objective:

$$\mathcal{L}_{QRC} = \mathcal{L}_{TD} - \lambda \mathcal{J}_{MI}, \quad (9)$$

where $\mathcal{L}_{TD}$ trains the Q-MIX value function, $\mathcal{J}_{MI}$ trains the receiver-specific message mask, and λ balances the two terms.

3) *Decentralized GNN: DDPG for Continuous Phase Split Control*: The Decentralized GNN model differs from CoLight and QRC-TSC because it does not select one discrete signal phase. Instead, each agent outputs continuous green time ratios for a fixed signal cycle. Therefore, the policy learning problem becomes continuous control, and DDPG is more suitable than discrete Q-learning.

The joint policy is factorized into decentralized policies:

$$\pi(s(t)) = \prod_{i \in N} \pi_i(s_i(t)). \quad (10)$$

After message passing, the graph-based state representation of agent i is denoted as $v_i(t)$. The actor network maps this representation to a continuous signal-timing action:

$$a_i(t) = \mu_i(v_i(t); \theta^\mu). \quad (11)$$

The critic network is used during training to evaluate whether the actor's continuous action improves the expected service rate (3). In other words, the actor learns how to output phase split decisions, while the critic provides the learning signal for improving these decisions. The GNN is trained together with the actor-critic framework, so the policy is learned from graph-based traffic representations rather than flattened local observations.

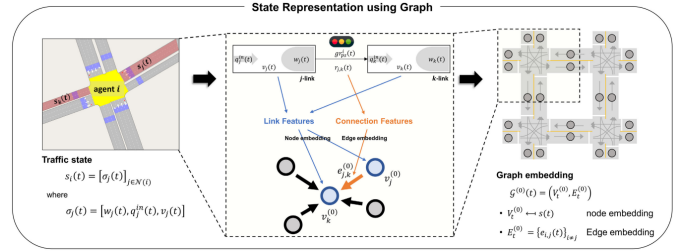


Fig. 4. State representation of decentralized GNN

This method is suitable for the Decentralized GNN model because phase split optimization requires continuous actions. If green time ratios were discretized, the action space would grow rapidly with the number of phases. DDPG allows the actor to directly output continuous green time ratios, while the critic evaluates the long-term effect of these continuous control decisions. The message-passing GNN further supports transferability by encoding spatial traffic correlations into a graph-based state representation.

4) *CGB-MATSC: Q-learning-style Regional Value Learning*: CGB-MATSC uses CGB-MAQL, a model-specific value-based MARL method that adapts Q-learning-style value estimation to cooperative regional control, as shown in Figure 5. This choice is suitable because each regional agent selects from a discrete set of high-level control modes, and learning action values allows the agent to directly compare different regional strategies according to their expected cumulative rewards.

The learned value function is defined as $Q_i(S_i, a_i)$, where S_i is the k -nearest-neighbor-based joint state and a_i denotes a high-level regional control action. The policy is then obtained by selecting the action with the highest estimated value:

$$\pi_i(S_i) = \arg \max_{a_i} Q_i(S_i, a_i). \quad (12)$$

The Q-learning-style update is:

$$Q_i(S_i, a_i) \leftarrow Q_i(S_i, a_i) + \alpha [r_i^{joint} + \gamma \max_{a'_i} Q_i(S'_i, a'_i) - Q_i(S_i, a_i)], \quad (13)$$

where α is the learning rate, γ is the discount factor, and S'_i is the next joint state. This update evaluates whether the selected

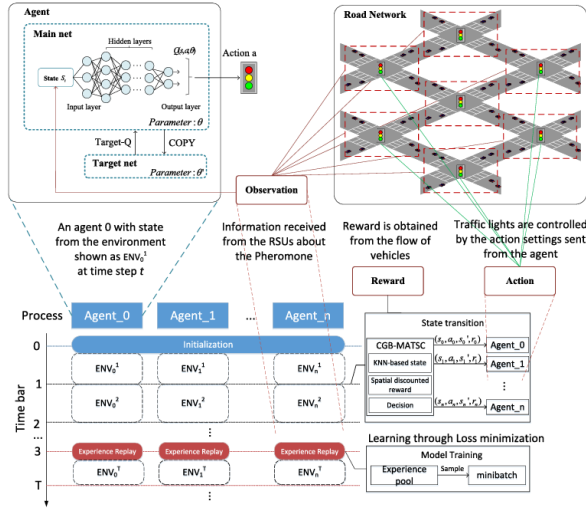


Fig. 5. Configuration of CGB-MATSC in the MARL framework.

regional action improves both the current traffic condition and the expected future value.

In this update, cooperation is introduced through both the state input and the reward term. The state S_i is constructed from the k -nearest-neighbor regional context, so the Q-function does not evaluate a high-level action based only on one isolated region. Meanwhile, the joint reward term r_i^{joint} , as introduced in the problem formulation, incorporates spatially discounted feedback from neighboring regions. Therefore, the Q-learning update guides each regional agent to select actions that improve not only its own local traffic condition, but also the cooperative performance of nearby regions.

5) *Summary of Policy Learning Choices*: Overall, the policy-learning choices are mainly determined by action-space design and coordination structure. CoLight and QRC-TSC use value-based methods for discrete phase selection, with QRC-TSC further using Q-MIX to align local decisions with a global objective. Decentralized GNN adopts DDPG because its action is continuous green-time ratio. CGB-MATSC uses a Q-learning-style regional value update because it selects high-level regional control strategies.

C. Cross-model Comparison of Communication and Coordination

The four models differ fundamentally in how they define communication and coordination, which can be summarized along two axes: *what is exchanged* and *how coordination is structured*.

a) *Communication mechanisms*: CoLight exchanges attention-weighted neighbor embeddings, implicitly prioritizing influential neighbors. QRC-TSC exchanges selectively masked messages, learning which bits to send and to whom. Decentralized GNN exchanges graph-structured features via multi-hop message passing along road links. CGB-MATSC exchanges regional-level pheromone signals through a hierarchical vehicle–infrastructure architecture.

b) *Coordination strategies*: CoLight coordinates by enriching each agent’s local observation with dynamic neighbor attention. QRC-TSC uses value decomposition (Q-MIX) during centralized training and selective communication during execution. Decentralized GNN embeds coordination into the graph representation itself, where message passing naturally propagates spatial traffic information. CGB-MATSC abstracts multiple intersections into regional agents, coordinating through spatially discounted rewards and high-level green-wave strategy selection.

c) *Summary*: These models represent a progression from intersection-level attention (CoLight), to learnable selective messaging (QRC-TSC), to link-level graph propagation (Decentralized GNN), and finally to regional role abstraction (CGB-MATSC). Each design prioritizes a different objective: CoLight handles dynamic neighbor influence, QRC-TSC reduces communication overhead, Decentralized GNN enables policy transferability, and CGB-MATSC achieves large-scale regional scalability. No single model excels in all dimensions; the choice depends on the specific network size, traffic variability, and communication constraints of the target deployment.

IV. EXPERIMENTAL RESULTS & COMPARISONS

This section compares the experimental results of the four selected MARL-based traffic signal control models from three perspectives: control performance, scalability and efficiency, and policy transferability. Since the four studies are evaluated under different simulators, road-network scales, traffic demand settings, baseline methods, and evaluation metrics, their raw numerical results are not directly comparable across papers. Therefore, different comparison strategies are used for different dimensions. For control performance, we mainly compare the relative improvement reported within each study, focusing on how each model improves over its own baselines. For scalability and efficiency, and policy transferability, we conduct evidence-based cross-study comparisons by examining whether each model provides support for large-scale deployment, reduced computational or communication cost, and generalization to unseen scenarios.

A. Control Performance

Across the four models, the control-performance metrics can be grouped into three categories: time cost that measures vehicle time loss, traffic conditions that describe the state of the road network, and RL metrics that verify the effectiveness of algorithm collaboration.

CoLight: Compared methods include traditional transportation (Fixedtime, MaxPressure) and RL-based approaches (CGRL, Individual RL, OneModel, Neighbor RL, GCN), plus two CoLight variants. From the perspective of time-cost and traffic conditions, CoLight achieves the best overall traffic-control performance on both synthetic and real-world networks, including average travel time and convergence speed. On the synthetic bidirectional 6×6 grid, CoLight reduces the average travel time to 170.11 s, which is about 12.8% lower

than MaxPressure (194.96 s) and about 29.9% lower than OneModel (242.63 s). On the large-scale New York network (196 intersections), CoLight achieves 1459.28 s, outperforming MaxPressure (1633.41 s) and GCN (1876.37 s) by clear margins. Its training time is also significantly lower than that of CGRL, Individual RL, and Neighbor RL, indicating lower computational cost.

The coordination-related results further show that effective cooperation depends on dynamic attention rather than fixed neighbor indexing. CoLight uses a Graph Attention Network (GAT) and this mechanism keeps model complexity independent of the number of intersections, allowing CoLight to scale to hundreds of traffic signals. Ablation studies reveal that the optimal neighbor number is 5 and 5 attention heads yield the best performance. Attention analysis reveals that upstream and arterial intersections receive higher spatial attention, and temporal attention adapts to traffic fluctuations, demonstrating that the model learns to focus on the most influential neighbors without requiring fixed ordering or per-pair message passing.

QRC-TSC: From the perspective of time-cost and traffic conditions, QRC-TSC achieves the best overall traffic-control performance on both networks, including mean waiting time, mean queue length, and mean speed. On the 4×4 grid network, it reduces the mean waiting time to 16.97 s/veh, which is about 54.4% lower than the strongest competing baseline. Its mean queue length is also about 64.8% lower than the best non-QRC baseline. On the Pasubio network, the improvement margin becomes smaller due to the more irregular real-world topology, but QRC-TSC still maintains the best overall performance.

The coordination-related results further show that effective communication depends on information quality rather than communication quantity. QRC-TSC occupies only 47.37% of the communication channel on the grid network and 63.41% on the Pasubio network, which is remarkably lower than the nearly 100% communication overhead incurred by full-communication methods. These results indicate that QRC-TSC can reduce vehicle time cost, relieve congestion, and improve traffic flow under both synthetic and real-world traffic conditions by learning a communication policy that decides which information should be sent and which agents should receive it.

Decentralized GNN: Baseline methods included fixed-time, random, longest-queue-first, and Webster control (with four variants of historical data accumulation). From the perspective of time-cost and traffic conditions, Decentralized GNN achieves substantial improvements on the real-world Gangnam network. After 1,000 training episodes, it reduces total travel time by 20.29%, average delay by 25.63%, and maximum queue length by 20.84%, while increasing space-mean speed by 5.90%. In a synthetic scenario with an abrupt demand pattern change, the model achieves 2.98% higher cumulative trip completion than the best Webster variant, demonstrating effective congestion relief and flow coordination under dynamic conditions.

The coordination-related results further show that effective

cooperation depends on bidirectional graph representation and message-passing rather than simple neighbor concatenation. A message-passing GNN iteratively updates node and edge features over multiple hops, enabling agents to sense traffic conditions several intersections away without explicit attention weights. This structural design captures fine-grained spatial correlations and supports policy transferability, making the model robust to unseen demand patterns.

CGB-MATSC is evaluated in SUMO on a synthetic network, Monaco and Harbin, and is compared with fixed-time control, CGB-Greedy, CGB-IQL and CGB-MA2C. Its advantage is most evident in the large-scale Harbin scenario, where CGB-MAQL achieves the best results in both time-cost and traffic-condition metrics, including average/peak waiting time and average/peak pheromone intensity. From the RL-control perspective, CGB-MAQL also reaches the shortest ending time in the synthetic scenario, indicating that its learned policy can discharge traffic more efficiently.

However, CGB-MAQL does not dominate all metrics in smaller or less coordination-demanding scenarios such as Monaco and the synthetic network. These results suggest a scale-dependent advantage: the regional MEC-based design is especially effective when traffic control depends on coordination across larger areas, but it is less uniformly superior when short-range traffic reactions are sufficient. Therefore, CGB-MATSC is stronger at large-scale regional flow coordination than at optimizing every local or peak congestion condition, and simpler greedy or actor-critic baselines may remain competitive on specific metrics.

Overall, the experimental results show that the four models improve MARL-based traffic signal control through different communication-driven mechanisms. CoLight improves travel-time performance by using attention to capture dynamically influential neighboring intersections, while QRC-TSC reduces congestion with selective communication, showing that useful information exchange can be more effective than full communication. Decentralized GNN improves delay, queue length, speed, and trip completion by embedding traffic interactions into graph-based message passing and continuous phase-split control. CGB-MATSC performs better in large-scale regional scenarios by shifting coordination from individual intersections to MEC-based regional agents. These findings suggest that model performance depends not only on the RL algorithm itself, but also on whether the communication structure matches the network scale, topology, and coordination requirements.

B. Scalability and Efficiency

Scalability and efficiency examine how well a model can handle increasing network sizes (e.g., number of intersections) and the computational or communication overhead it incurs. Key considerations include parameter complexity, training time, communication bandwidth usage, and the ability to be deployed on large-scale road networks. This subsection compares the four models along these dimensions.

CoLight achieves high scalability through parameter sharing and index-free attention aggregation. Its space and time complexity are both $O(m^2L)$ —independent of the number of intersections—which enables training on networks with up to 196 intersections. Empirically, its training time is comparable to that of simple shared-model baselines (e.g., OneModel) and significantly lower than individual-agent RL. Communication overhead is not explicitly measured, but the attention mechanism aggregates neighbor information without per-pair message passing.

QRC-TSC shows strong communication efficiency but limited evidence for city-level scalability. Its main advantage lies in communication bandwidth usage: agents use only about 47%–65% of the communication channel while maintaining strong control performance. This suggests that selective communication can reduce unnecessary message passing without weakening coordination. In terms of parameter complexity, the Q-MIX-based CTDE framework supports decentralized execution, but the additional communication policy also increases model complexity. The paper does not provide a direct runtime comparison, so its training efficiency is difficult to evaluate. Moreover, the experiments are limited to a 4×4 synthetic grid and one neighborhood-scale Pasubio network. Therefore, QRC-TSC is mainly scalable from the perspective of communication efficiency, while its deployment on larger city-level networks remains insufficiently verified.

Decentralized GNN’s complexity scales linearly with the number of edges. However, it has only been validated on a relatively small network (8 internal intersections plus 16 boundary intersections, totalling 24). Training uses CTDE with DDPG, but training time or convergence speed is not directly compared against other MARL methods. Communication takes the form of message-passing GNN updates over multiple hops; no specialized mechanism for reducing communication bandwidth is introduced, and its scalability to networks with hundreds of intersections remains untested.

CGB-MATSC provides strong scalability evidence through its validation on the large-scale Harbin scenario, which covers a geographic area of $6000m \times 7000m$ and contains 341 intersections. In this scenario, CGB-MAQL achieves the best results among the compared methods in both average waiting time and average road pheromone, suggesting that the model can maintain effective control performance as the network scale increases. This strong performance can be attributed to its regional abstraction, which reduces the difficulty of dense intersection-level modeling, while pheromone-based green-wave control and spatially discounted reward help coordinate traffic flows across larger urban areas. Therefore, CGB-MATSC is more suitable for city-scale traffic signal control than methods whose scalability is only validated on small or medium-sized networks. Although communication efficiency is not its primary contribution, the hierarchical communication structure and compact high-level action space still help reduce the burden of dense intersection-to-intersection coordination.

In summary, CoLight and CGB-MATSC demonstrate strong scalability, each through a different strategy: CoLight

via parameter sharing and constant model complexity, and CGB-MATSC through regional abstraction that decouples the number of agents from the number of intersections. QRC-TSC focuses on communication efficiency, using selective messaging to reduce channel usage, but its experimental validation is limited to small networks. The Decentralized GNN model, despite its linear per-edge complexity, has not been tested at scale and lacks evidence of scalability beyond a modest 24-intersection scenario. Thus, clear differences exist in how each model addresses the scalability challenge, with CoLight and CGB-MATSC being the most suitable for very large road networks.

C. Policy Transferability

Policy transferability refers to a model’s ability to generalize to unseen traffic scenarios (e.g., reversed demand, different network topologies). It is critical for real-world deployment under non-stationary conditions but is often overlooked in MARL-TSC literature.

CoLight does not evaluate transferability; its robustness to changing demand remains unknown.

QRC-TSC also does not explicitly test transferability. While its communication mechanism is validated under different settings, direct policy application to new traffic situations without retraining is not examined.

Decentralized GNN is the only method that systematically evaluates transferability. Trained on AM peak (residential→CBD) and tested on unseen PM peak (CBD→residential), the pretrained policy maintains higher speed and trip completion than Webster control, despite a slight density increase. This demonstrates that its bidirectional graph representation and message-passing GNN enable adaptation to reversed demand.

CGB-MATSC provides moderate evidence: the algorithm is trained on a synthetic network and then tested on larger city-scale scenarios (Monaco with 324 intersections, Harbin with 341 intersections). This shows cross-scale and cross-layout portability, but it does not specifically examine demand-pattern shifts (e.g., reversed peak flows). Thus, its transferability is weaker and less targeted than that of Decentralized GNN.

In summary, only Decentralized GNN offers rigorous transferability evaluation via demand reversal, while CGB-MATSC gives partial evidence for network-scale transfer. CoLight and QRC-TSC do not address this dimension. Future work should prioritize cross-scenario generalization for practical adaptive control under non-stationary traffic.

V. OPEN RESEARCH CHALLENGES

Despite the promising performance of Multi-Agent Reinforcement Learning (MARL) in traffic signal control, several critical challenges remain to be addressed before these models can be transitioned from experimental environments to real-world infrastructure.

A. Bridging the Simulation-to-Real (Sim-to-Real) Gap

As highlighted in the survey [2], the vast majority of current research relies exclusively on simulators. While studies like Decentralized GNN [5] have begun addressing policy transferability across different network topologies, a significant gap persists regarding real-world deployment.

- **Realism of Environments:** Current models often overlook human intervention and predictable extreme scenarios (e.g., accidents or road construction). Future research should integrate more realistic variables into simulators to ensure that agents, such as those in Adaptive CGB-MATSC, do not converge on local optima that fail under real-world volatility.
- **Zero-shot Generalization:** As noted in the analysis of Yoon et al. [5], the ability to deploy a pre-trained model on an entirely unseen city grid without extensive retraining remains an open frontier.

B. Standardization of Benchmarking and Comparisons

A major bottleneck in the field is the lack of a unified evaluation framework.

- **Advanced Baselines:** Many studies still use simple DQN or fixed-time controllers as primary baselines. However, new MARL approaches should be rigorously compared against existing multi-agent distributed control and coordinated control methods.
- **Integration with Traditional Systems:** There is a notable lack of comparison with actuated controllers, which are currently the most prevalent real-world implementation. Future studies must demonstrate that RL-based solutions provide a significant marginal gain over these established systems.

C. Robustness under Partial Observability and Noise

The definition of “state” lacks a universal standard, and most models assume perfect data acquisition.

- **Noisy and Incomplete Data:** Real-world sensors frequently malfunction or provide noisy data. While QRC-TSC focuses on communication efficiency, it assumes stable message passing. A vital research direction lies in designing agents that remain resilient when communication channels are interrupted or when specific detection devices fail.
- **Dynamic Coordination:** The fixed geographic grouping in CGB-MATSC and the static graph structures in CoLight may struggle with dynamic traffic shifts. Future work should explore “Dynamic Grouping” or “Multi-modal Attention” that can adaptively reconfigure coordination patterns based on real-time sensor reliability and heterogeneous data sources.

VI. CONCLUSION

This literature research has synthesized the evolving landscape of Multi-Agent Reinforcement Learning (MARL) specifically through the lens of communication mechanisms in

Traffic Signal Control (TSC). By critically evaluating QRC-TSC [1], CGB-MATSC [6], Decentralized GNN [5], and CoLight [4], we demonstrate that advanced communication, ranging from representational embedding sharing to graph-based spatial modeling, is the cornerstone of transitioning from localized intersections to coordinated urban networks. These methodologies effectively mitigate the constraints of partial observability, allowing agents to capture long-range spatio-temporal dependencies that traditional actuated systems often overlook.

Our analysis reveals a diverse array of objectives among these methodologies, suggesting that no single approach serves as a universal solution. For instance, CoLight leverages GAT to move beyond traditional pre-calculated offsets, enabling dynamic spatial coordination that adapts to real-time neighbor influence. In contrast, QRC-TSC prioritizes communication efficiency by employing a selective masking mechanism to mitigate noise, redundancy, and bandwidth waste inherent in full-communication networks. To enhance policy transferability across varying road layouts, the Decentralized GNN utilizes bidirectional graph representations and multi-hop message passing to capture link-level spatial correlations. Finally, to address city-level scalability, CGB-MATSC abstracts individual intersections into regional units using KNN joint states and spatially discounted rewards to encourage group-based cooperation. These distinct priorities highlight a fundamental trade-off between coordination granularity, communication cost and network scale.

Despite these advancements, our survey identifies a persistent “reality gap” between high-fidelity simulators and the stochasticity of physical infrastructure. To bridge this divide, future research should not only focus on algorithmic efficiency but also on communication robustness under constrained bandwidth and interoperability with legacy traffic systems. Furthermore, integrating heterogeneous data (e.g., connected vehicle trajectories and pedestrian flows) will be essential. Moving forward, the goal is to refine these MARL frameworks into resilient, interpretable, and scalable solutions capable of orchestrating the next generation of intelligent urban mobility.

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